

Hierarchy of infinities



Outline

Comparison of infinities for functions



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Introduction: comparison of infinities for functions

- ▶ In this section we compare the **speed** at which three families of functions tend to $+\infty$: **logarithms**, **powers** and **exponentials**
- ▶ The result is a true **hierarchy**: $\text{logarithms} \ll \text{powers} \ll \text{exponentials}$
- ▶ With a change of variable we will obtain many useful limits also as $x \rightarrow 0^+$ and as $x \rightarrow -\infty$, which allow us to solve indeterminate forms of the type $0 \cdot \infty$ and $\frac{\infty}{\infty}$

Comparison of infinities

Theorem 1: Comparison of infinities

Given $\alpha, \beta, \lambda > 0$, $a, b > 1$, we have

$$\lim_{x \rightarrow +\infty} \frac{\log_a^\beta x}{x^\alpha} = 0 \quad \text{and} \quad \lim_{x \rightarrow +\infty} \frac{x^\alpha}{b^{\lambda x}} = 0$$

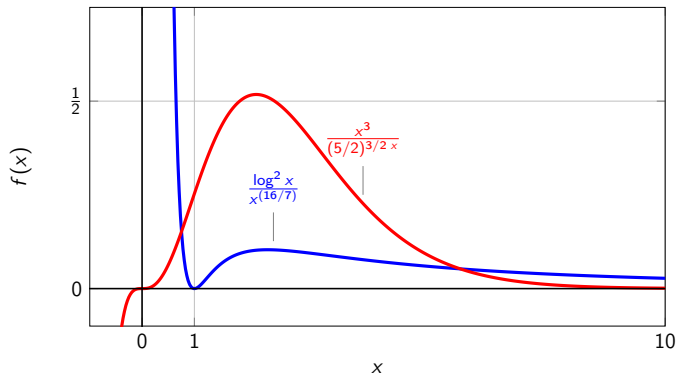
Proof: the theorem will be proved with the tools of differential calculus.

Hierarchy of infinities

- ▶ Any power with positive exponent is an infinity of higher order than any power of logarithms with base > 1 , and
- ▶ Any exponential with base > 1 is an infinity of higher order than any power with positive exponent.

Example

$$\lim_{x \rightarrow +\infty} \frac{\log^2 x}{x^{(16/7)}} = 0, \quad \lim_{x \rightarrow +\infty} \frac{x^3}{(5/2)^{3/2} x} = 0$$



Comparison of infinities with a function $\eta(x)$

If a function $\eta(x) \rightarrow +\infty$ as $x \rightarrow c \in \mathbb{R}^*$, we have, from the comparison of infinities theorem:

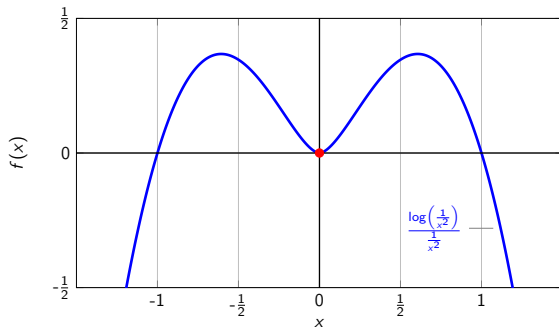
$$\lim_{x \rightarrow c} \frac{\left(\log_a (\eta(x)) \right)^\beta}{(\eta(x))^\alpha} = 0 \quad \text{and} \quad \lim_{x \rightarrow c} \frac{(\eta(x))^\alpha}{b^\lambda \eta(x)} = 0$$

with $\alpha, \beta, \lambda > 0, a, b > 1$

Example

$$\lim_{x \rightarrow 0} \frac{\log\left(\frac{1}{x^2}\right)}{\frac{1}{x^2}} = 0$$

since, with $\eta(x) = \frac{1}{x^2}$, we have $\eta(x) \rightarrow +\infty$ as $x \rightarrow 0$.



With the change of variable $t = \frac{1}{x}$ we now turn limits as $x \rightarrow +\infty$ into limits as $t \rightarrow 0^+$: in this way we obtain a series of consequences of the theorem.

Powers and logarithms at 0^+

Observation 1

$$\lim_{x \rightarrow 0^+} x^\alpha \log_a^\beta x = 0$$

$$\left(\alpha > 0, \beta = \frac{m}{n}, n, m \in \mathbb{N}, n \text{ odd}, a > 1 \right)$$

Proof

Given $\alpha, \beta > 0, a > 1$ we have:

$$\lim_{x \rightarrow +\infty} \frac{\log_a^\beta x}{x^\alpha} = 0 \quad \text{and} \quad \frac{\log_a^\beta x}{x^\alpha} = \left(\frac{1}{x}\right)^\alpha \left(-\log_a \left(\frac{1}{x}\right)\right)^\beta$$

Setting $t = \frac{1}{x}$, $x \rightarrow +\infty$ is equivalent to $t \rightarrow 0^+$, and substituting we have:

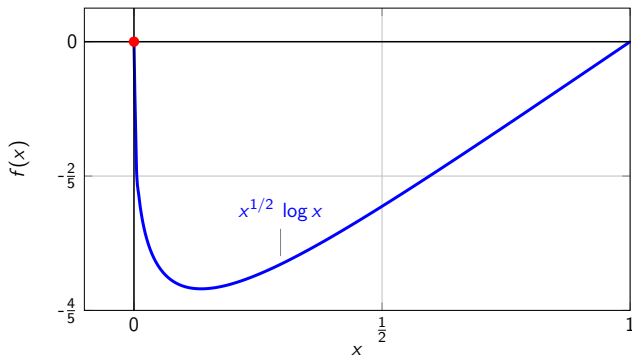
$$\lim_{x \rightarrow +\infty} \frac{\log_a^\beta x}{x^\alpha} = (-1)^\beta \lim_{t \rightarrow 0^+} t^\alpha \log_a^\beta t = 0$$



Example (1/2)

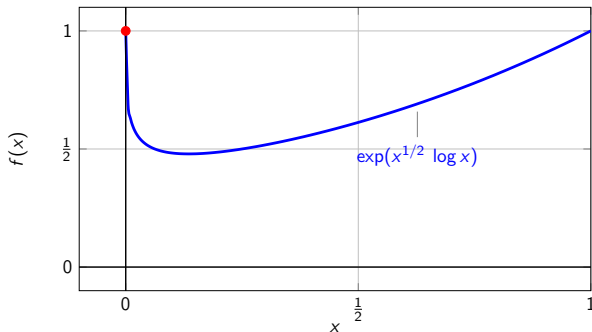
$$\lim_{x \rightarrow 0^+} x^{1/2} \log x = 0^-$$

since we have $\alpha = \frac{1}{2} > 0$, $\beta = 1 > 0$ and $a = e > 1$.



Example (2/2)

$$\lim_{x \rightarrow 0^+} x^{\sqrt{x}} = 1^- \quad \text{since} \quad \lim_{x \rightarrow 0^+} x^{\sqrt{x}} = \lim_{x \rightarrow 0^+} \underbrace{e^{x^{1/2} \log x}}_{=\exp(x^{1/2} \log x)} = e^{0^-} = 1^-$$



Powers and logarithms with a function $\varepsilon(x)$

If a function $\varepsilon(x)$ tends to 0^+ as $x \rightarrow c \in \mathbb{R}^*$, it follows from the previous remark that:

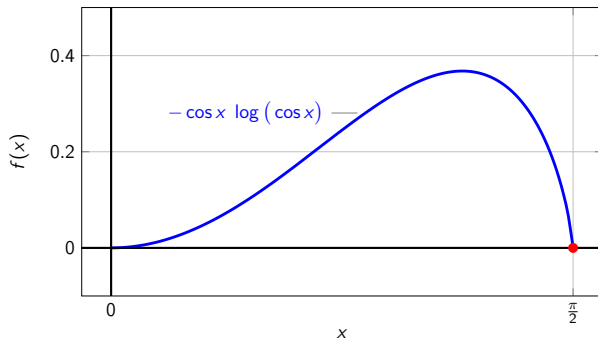
$$\lim_{x \rightarrow c} (\varepsilon(x))^\alpha (\log_a (\varepsilon(x)))^\beta = 0$$

$$\left(\alpha > 0, \beta = \frac{m}{n}, n, m \in \mathbb{N}, n \text{ odd}, a > 1 \right)$$

Example

$$\lim_{x \rightarrow \frac{\pi}{2}^-} -\cos x \log(\cos x) = 0^+$$

since, with $\varepsilon(x) = \cos x$, we have $\varepsilon(x) \rightarrow 0^+$ as $x \rightarrow \frac{\pi}{2}^-$ and $\alpha = \beta = 1$.



Limits of the reciprocals

From the comparison of infinities theorem it follows that:

$$\lim_{x \rightarrow +\infty} \frac{x^\alpha}{\log_a^\beta x} = +\infty \quad \text{and} \quad \lim_{x \rightarrow +\infty} \frac{b^{\lambda x}}{x^\alpha} = +\infty$$

$$\text{with } \alpha, \beta, \lambda > 0, \quad a, b > 1$$

Powers and exponentials at 0^+ (1/2)

Observation 2

$$\lim_{x \rightarrow 0^+} x^\alpha b^{\lambda/x} = +\infty \quad (\alpha, \lambda > 0, b > 1)$$

Proof

Given $\alpha, \lambda > 0, b > 1$, we have:

$$\lim_{x \rightarrow +\infty} \frac{b^{\lambda x}}{x^\alpha} = +\infty$$

Setting $t = \frac{1}{x}$, $x \rightarrow +\infty$ is equivalent to $t \rightarrow 0^+$, and substituting we have:

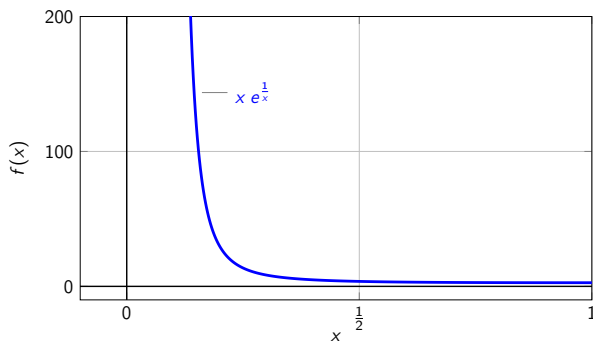
$$\lim_{x \rightarrow +\infty} \frac{b^{\lambda x}}{x^\alpha} = \lim_{x \rightarrow +\infty} \left(\frac{1}{x}\right)^\alpha b^{\lambda x} = \lim_{t \rightarrow 0^+} t^\alpha b^{\lambda/t} = +\infty$$



Example

$$\lim_{x \rightarrow 0^+} x e^{\frac{1}{x}} = +\infty$$

since we have $\alpha = \lambda = 1$ and $b = e > 1$



Powers and exponentials at 0^+ (2/2)

Observation 3

$$\lim_{x \rightarrow 0^+} x^{-\alpha} b^{-\lambda/x} = 0 \quad (\alpha, \lambda > 0, b > 1)$$

Proof

Given $\alpha, \lambda > 0, b > 1$ we have:

$$\lim_{x \rightarrow +\infty} \frac{x^\alpha}{b^{\lambda x}} = 0$$

Setting $t = \frac{1}{x}$, $x \rightarrow +\infty$ is equivalent to $t \rightarrow 0^+$, and substituting we have:

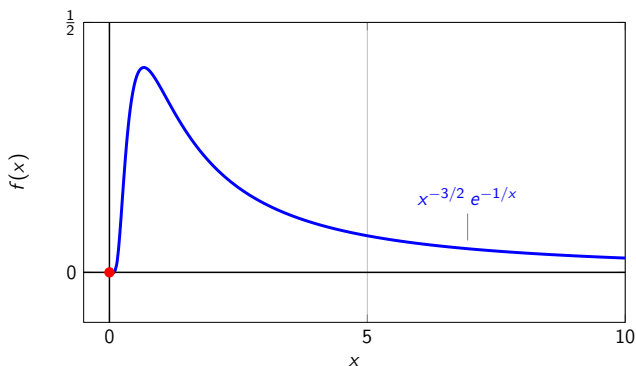
$$\lim_{x \rightarrow +\infty} \frac{x^\alpha}{b^{\lambda x}} = \lim_{x \rightarrow +\infty} \left(\frac{1}{x} \right)^{-\alpha} \frac{1}{b^{\lambda x}} = \lim_{t \rightarrow 0^+} t^{-\alpha} b^{-\lambda/t} = 0$$



Example

$$\lim_{x \rightarrow 0^+} x^{-3/2} e^{-1/x} = 0$$

since we have $\alpha = \frac{3}{2}$, $\lambda = 1$ and $b = e > 1$.



Finally, with the change of variable $t = -x$ we turn limits as $x \rightarrow +\infty$ into limits as $t \rightarrow -\infty$.

Powers and exponentials at $-\infty$

Observation 4

$$\lim_{x \rightarrow -\infty} x^\alpha b^{\lambda x} = 0$$

$$\left(\alpha = \frac{m}{n} > 0, n, m \in \mathbb{N}, n \text{ odd}, m \neq 0, \lambda > 0, b > 1 \right)$$

Proof

Given $\alpha, \lambda > 0, b > 1$ we have:

$$\lim_{x \rightarrow +\infty} \frac{x^\alpha}{b^{\lambda x}} = 0$$

Setting $t = -x$, $x \rightarrow +\infty$ is equivalent to $t \rightarrow -\infty$, and substituting we have:

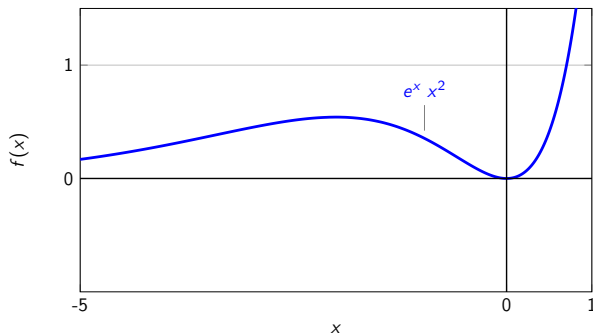
$$\lim_{x \rightarrow +\infty} \frac{x^\alpha}{b^{\lambda x}} = \lim_{t \rightarrow -\infty} \frac{(-t)^\alpha}{b^{-\lambda t}} = (-1)^\alpha \lim_{t \rightarrow -\infty} t^\alpha b^{\lambda t} = 0$$



Example

$$\lim_{x \rightarrow -\infty} x^2 e^x = 0$$

since we have $\alpha = 2$, $\lambda = 1$ and $b = e > 1$.



Summary (1/2)

Hierarchy of infinities (as $x \rightarrow +\infty$, with $\alpha, \beta, \lambda > 0$, $a, b > 1$):

- ▶ $\log_a^\beta x \ll x^\alpha \ll b^{\lambda x}$: logarithms grow more slowly than powers, powers more slowly than exponentials
- ▶ The same limits hold replacing x with a function $\eta(x) \rightarrow +\infty$
- ▶ The reciprocals tend to $+\infty$

Powers and logarithms at 0^+ :

- ▶ $x^\alpha \log_a^\beta x \rightarrow 0$ as $x \rightarrow 0^+$ (also with $\varepsilon(x) \rightarrow 0^+$ in place of x)

Summary (2/2)

Powers and exponentials:

- ▶ $x^\alpha b^{\lambda/x} \rightarrow +\infty$ and $x^{-\alpha} b^{-\lambda/x} \rightarrow 0$ as $x \rightarrow 0^+$
- ▶ $x^\alpha b^{\lambda x} \rightarrow 0$ as $x \rightarrow -\infty$

Method:

- ▶ All these consequences are obtained from the comparison of infinities theorem with the changes of variable $t = \frac{1}{x}$ and $t = -x$
- ▶ They allow us to solve indeterminate forms of the type $0 \cdot \infty$ and $\frac{\infty}{\infty}$